Nifty100 Financial Intelligence Platform

Analyst Guide

1. Introduction

The Nifty100 Financial Intelligence Platform is a complete financial analytics solution developed using Python, SQLite, FastAPI and Streamlit. It provides company analysis, valuation, screening, clustering, peer comparison and portfolio analytics.

2. System Requirements

Python 3.14+, SQLite, Streamlit, FastAPI, Pandas, Plotly and ReportLab.

3. Installation

Install all dependencies using requirements.txt and create a virtual environment before execution.

4. Database

The platform uses db/nifty100.db containing financial information for 92 companies.

5. ETL

Execute the ETL pipeline to load and validate company financial statements.

6. Analytics

Generate KPIs, CAGR, clustering, correlation matrix, outlier reports and portfolio statistics.

7. Dashboard

The Streamlit dashboard provides Company Profile, Screener, Peer Comparison, Sector Intelligence, Trends and Reports.

8. FastAPI

The REST API exposes company data, ratios, screener, sectors, peers, valuation and documents.

9. Reports

Reports include executive summaries, portfolio summaries, valuation summaries and company tearsheets.

10. Testing

All automated tests are executed using pytest. The project currently passes all available tests.

11. Performance

Concurrent API load testing was successfully completed within the Sprint target.

12. Outputs

Generated outputs include cluster labels, outlier reports, portfolio statistics, heatmaps and OpenAPI specification.

13. Troubleshooting

Verify database path, dependencies and API server status if issues occur.

14. Best Practices

Always execute ETL before analytics. Regenerate reports after refreshing the database.

15. Conclusion

The project provides a complete financial intelligence platform suitable for equity research, screening, clustering and reporting.